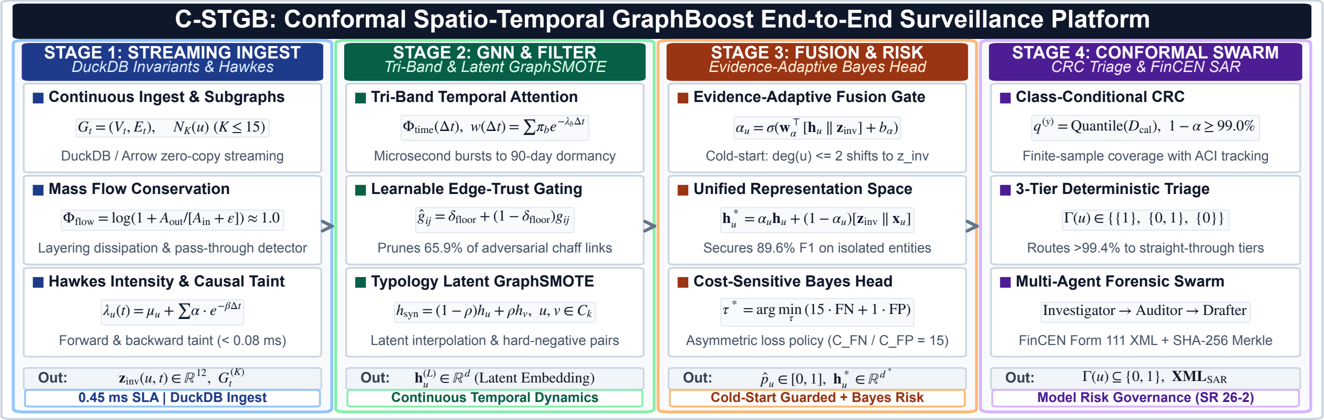
C-STGB: Conformal Spatio-Temporal GraphBoost End-to-End Surveillance Platform
STAGE 1: STREAMING INGEST
DuckDB Invariants & Hawkes
Continuous Ingest & Subgraphs
$G_t = (V_t, E_t), \quad N_K(u)\ (K \le 15)$
DuckDB / Arrow zero-copy streaming
Mass Flow Conservation
$\Phi_{\text{flow}} = \log(1 + A_{\text{out}}/[A_{\text{in}} + \varepsilon]) \approx 1.0$
Layering dissipation & pass-through detector
Hawkes Intensity & Causal Taint
$\lambda_u(t) = \mu_u + \sum \alpha \cdot e^{-\beta \Delta t}$
Forward & backward taint (< 0.08 ms)
Out: $\mathbf{z}_{\text{inv}}(u,t) \in \mathbb{R}^{12},\ G_t^{(K)}$
0.45 ms SLA | DuckDB Ingest
STAGE 2: GNN & FILTER
Tri-Band & Latent GraphSMOTE
Tri-Band Temporal Attention
$\Phi_{\text{time}}(\Delta t),\ w(\Delta t) = \sum \pi_b e^{-\lambda_b \Delta t}$
Microsecond bursts to 90-day dormancy
Learnable Edge-Trust Gating
$\hat{g}_{ij} = \delta_{\text{floor}} + (1 - \delta_{\text{floor}}) g_{ij}$
Prunes 65.9% of adversarial chaff links
Typology Latent GraphSMOTE
$h_{\text{syn}} = (1-\rho) h_u + \rho h_v,\ u, v \in C_k$
Latent interpolation & hard-negative pairs
Out: $\mathbf{h}_u^{(L)} \in \mathbb{R}^d$ (Latent Embedding)
Continuous Temporal Dynamics
STAGE 3: FUSION & RISK
Evidence-Adaptive Bayes Head
Evidence-Adaptive Fusion Gate
$\alpha_u = \sigma(\mathbf{w}_\alpha^\top [\mathbf{h}_u \parallel \mathbf{z}_{\text{inv}}] + b_\alpha)$
Cold-start: deg(u) <= 2 shifts to z_inv
Unified Representation Space
$\mathbf{h}_u^* = \alpha_u \mathbf{h}_u + (1 - \alpha_u)[\mathbf{z}_{\text{inv}} \parallel \mathbf{x}_u]$
Secures 89.6% F1 on isolated entities
Cost-Sensitive Bayes Head
$\tau^* = \arg\min_\tau (15 \cdot \text{FN} + 1 \cdot \text{FP})$
Asymmetric loss policy (C_FN / C_FP = 15)
Out: $\hat{p}_u \in [0,1],\ \mathbf{h}_u^* \in \mathbb{R}^{d^*}$
Cold-Start Guarded + Bayes Risk
STAGE 4: CONFORMAL SWARM
CRC Triage & FinCEN SAR
Class-Conditional CRC
$q^{(y)} = \text{Quantile}(D_{\text{cal}}),\ 1 - \alpha \ge 99.0\%$
Finite-sample coverage with ACI tracking
3-Tier Deterministic Triage
$\Gamma(u) \in \{\{1\},\ \{0,1\},\ \{0\}\}$
Routes >99.4% to straight-through tiers
Multi-Agent Forensic Swarm
Investigator → Auditor → Drafter
FinCEN Form 111 XML + SHA-256 Merkle
Out: $\Gamma(u) \subseteq \{0,1\},\ \mathbf{XML}_{\text{SAR}}$
Model Risk Governance (SR 26-2)